

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Pharm (Quality Assurance)

University Theory Examination May 2016

PH816.07 PHARMACEUTICAL VALIDATION AND DOCUMENTATION

Date: 06.05.2016, Friday

Time: 01.30 p.m. to 04.30 p.m.

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1** Define process validation. Discuss types of process validation. Describe scope and advantages of process validation **10**
- Q.2** Write a detail note on equipment validation. **10**

OR

- Q.2** Explain importance of cleaning validation. What are the different methods for cleaning in pharmaceutical manufacturing. Describe various steps for effective cleaning validation program. **10**
- Q.3** Answer the following (Any three) **15**
- (A) How the validation of autoclave can be performed? Describe.
- (B) What are the advantages of validation of HVAC system?
- (C) What are the acceptance criteria of validation of tablet compression machine?
- (D) What is the process of cleaning validation of double cone mixer?

SECTION-II

- Q.4** Define market complaint. What are the objectives, purpose and types of complaints? Describe procedure for handling market complaint and its relevant documentation. **10**
- Q.5** What is product recall? Give classification of product recalls. Write general procedure for product recall. Give outline of product recall documentation. **10**

OR

- Q.5** Explain Batch Manufacturing Record and Master Formula Record with respect to content/components. Outline BMR and MFR formats. Give differences between BMR and MFR. **10**
- Q.6** Write brief notes on followings: (Any three) **15**
- (A) Validation Master Plan
- (B) Good documentation practices in pharmaceutical industry
- (C) IQ, OQ, PQ Documentation
- (D) Paperless documentation system